Singular Value Decomposition

Hoa Vu
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Julia

Why Julia?

- Built for scientific computing. Used in industry and government (NASA, National Labs, ASML, IBM, Amazon, JP Morgan)
- Just-in-Time compilation. Very fast (can be close to C).
- Python-like ergonomics, less slowdown.
- Can be called directly from Python.
- Cons: Still not as popular as Python which you should already known.

Vectors, Matrices, System of Linear Equations

```
using LinearAlgebra

a = 0.85
n = 5
M = [
    0.0   1/5  0.0   0.0   0.0
    1.0   1/5  1/2   1/2   1.0
    0.0   1/5  0.0   1/2   0.0
    0.0   1/5  1/2   0.0   0.0
    0.0   1/5  0.0   0.0   0.0
]

b = (1 - a) / n .* ones(n)
pi = (I - a * M) \ b # Solve (I - a M) pi = b
```

Vectors, Matrices, System of Linear Equations

```
A * A              # matrix-matrix multiply
A * v              # matrix-vector multiply

transpose(A)       # transpose (no conjugation)

dot(v, v)          # dot product
norm(v)            # vector norm (2-norm by default)
tr(A)              # trace
det(A)             # determinant
```

What is Singular Value Decomposition (SVD)?

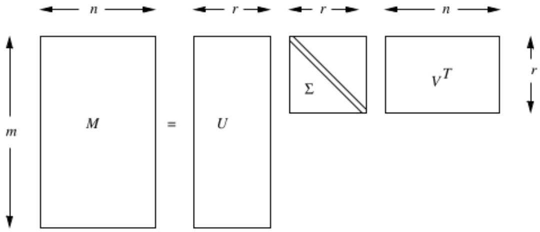

Figure 11.5: The form of a singular-value decomposition

Figure: SVD

What is Singular Value Decomposition (SVD)?

Definition

For any matrix $M \in \mathbb{R}^{m \times n}$ of rank r, there exist matrices:

$$M = U \Sigma V^T$$

- $U \in \mathbb{R}^{m \times r}$: orthonormal columns
- $V \in \mathbb{R}^{n \times r}$: orthonormal columns
- $\Sigma \in \mathbb{R}^{r \times r}$: diagonal (singular values)

Example

$$
\begin{array}{c c}
 & \begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Star Wars} & \text{Casablanca} & \text{Titanic} \end{array} \\
\begin{array}{r} \text{Joe} \\ \text{Jim} \\ \text{John} \\ \text{Jack} \\ \text{Jill} \\ \text{Jenny} \\ \text{Jane} \end{array} &
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 0 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 0 & 0 & 2 & 2
\end{bmatrix}
\end{array}
$$

Figure: Ratings of movies by users

Example

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 & 2 \end{bmatrix}}_{M} \approx \underbrace{\begin{bmatrix} .14 & 0 \\ .42 & 0 \\ .56 & 0 \\ .70 & 0 \\ 0 & .60 \\ 0 & .75 \\ 0 & .30 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix}}_{V^T}$$

Figure: rank-2 approximation for the matrix M (of the ratings example)

Singular Values

- Diagonal entries of Σ: $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$
- Measure the **importance** of each latent component
- Large $\sigma_i \Rightarrow$ more information

Rank

The number of nonzero singular values equals the rank of M.

Interpretation (Intuition)

- SVD finds hidden **concepts** in the data (i.e., romance, sci-fi genres)
- Rows of U: how rows of M relate to concepts
- Rows of V^T: how columns of M relate to concepts
- Σ: strength of each concept

Mental Model

$$\textit{Data} = \textit{(row weights)} \times \textit{(concept strength)} \times \textit{(column weights)}$$

Interpreting Σ: Concept Strength

Given the SVD

$$M = U \Sigma V^T,$$

- The decomposition discovers **latent concepts** (patterns) in the matrix.
- Σ is diagonal:

$$\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \dots, \sigma_r), \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

- Each singular value σ_i measures the **importance/strength** of concept i.
- Larger $\sigma_i \Rightarrow$ that concept explains more of the structure in M.

Columns of U and V Are the Concepts

SVD can be viewed as a sum of rank-1 "concept" components:

$$M = \sum_{i=1}^{r} \sigma_i \, u_i {v_i}^T,$$

where u_i is the ith column of U and v_i is the ith column of V.

- u_i: how each **row entity** (e.g., user) loads on concept i
- v_i: how each **column entity** (e.g., movie) loads on concept i
- σ_i: how strongly that concept contributes to reconstructing M

Why SVD is Useful

- Compact representation of large matrices
- Noise reduction
- Reveals structure in data
- Basis for recommender systems, PCA, NLP

Dimensionality Reduction

Idea

Keep only the top $k < r$ singular values:

$$M \approx U_k \Sigma_k V_k^T$$

- Drop smallest singular values
- Reduces storage and computation
- Preserves most important structure

Why Dropping Small Singular Values Works

- Error measured using Frobenius norm
- Truncated SVD gives the **best rank-k approximation**
- Minimizes reconstruction error

Energy Rule (Heuristic)

Keep enough singular values to retain $\sim$90% of:

$$\sum_i \sigma_i^2$$

Querying Using Concepts (SVD idea)

Assume we have an SVD (or truncated SVD) of the ratings matrix:

$$M \approx U\Sigma V^T$$

- Treat columns of V as a basis for a low-dimensional *concept space*.
- Map a user (row vector) from movie-space $\rightarrow$ concept-space using V.

Example: Map a New User into Concept Space

Quincy has rated only *The Matrix* with score 4:

$$q = [4, 0, 0, 0, 0]$$

Map Quincy to concept space by multiplying with V:

$$qV = [2.32, 0]$$

- Interpretation: strong interest in concept 1, none in concept 2.
- (Note: many texts display V^T; this step uses V.)

Predict Ratings by Mapping Back

Map concept-space vector back to movie space using V^T:

$$[2.32, 0]\, V^T = [1.35, 1.35, 1.35, 0, 0]$$

- Predicts Quincy would like movies aligned with the first concept (here: the first three movies), and not the last two.

Find Similar Users in Concept Space

Also map existing users into concept space (using V):

$$\text{Joe} \mapsto [1.74, 0], \qquad \text{Jill} \mapsto [0, 5.68]$$

Measure similarity via cosine distance (angle) in concept space:

- Quincy vs Joe: same direction $\Rightarrow$ cosine distance $= 0$
- Quincy vs Jill: orthogonal (dot product 0) $\Rightarrow$ angle 90°, cosine distance $= 1$